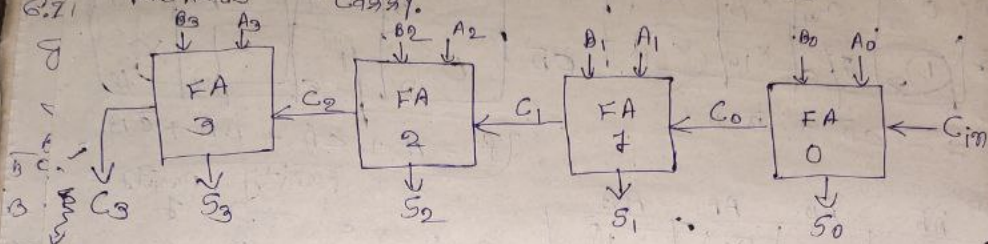


2-bit Comparators

★ Carry Look Ahead Adder (CLA Adder) :-

- CLA Adder is faster in terms of operation speed.
- In 4-bit parallel adder using full adders, propagation delay is introduced due to dependence on previous carry.



• Produce Generalised Function :-

⇒ Focus on Full Adder 1 & we will generate our function from it.

B_1	A_1	C_0	C_1
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

⇒ Thus,

$$C_1 = \bar{B}_1 A_1 C_0 + B_1 \bar{A}_1 C_0 + B_1 A_1 \bar{C}_0 + B_1 A_1 C_0$$

$$C_1 = B_1 A_1 (C_0 + \bar{C}_0) + C_0 (B_1 \bar{A}_1 + \bar{B}_1 A_1)$$

$$= B_1 A_1 + C_0 (B_1 \oplus A_1)$$

$$C_i = \underbrace{B_i A_i}_{\text{Carry Generator } G_i} + \underbrace{(B_i \oplus A_i)}_{\text{Carry Propagator } P_i} \cdot C_{i-1}$$

$$\therefore C_i = G_i + P_i \cdot C_{i-1}$$

↓
[Generalised Formula]

$$G_i = A_i B_i$$

$$P_i = A_i \oplus B_i$$

How to make it previous Carry Independent:

1. For $i = 0$;

$$C_0 = a_0 + p_0 \cdot C_{in}$$

①

2. For $i = 1$;

$$\begin{aligned} C_1 &= a_1 + p_1 \cdot C_0 \\ &= a_1 + p_1 (a_0 + p_0 \cdot C_{in}) \\ &= a_1 + p_1 a_0 + p_1 p_0 \cdot C_{in} \end{aligned}$$

3. For $i = 2$;

$$\begin{aligned} C_2 &= a_2 + p_2 \cdot C_1 \\ &= a_2 + p_2 (a_1 + p_1 a_0 + p_1 p_0 \cdot C_{in}) \\ C_2 &= a_2 + p_2 a_1 + p_2 p_1 a_0 + p_2 p_1 p_0 \cdot C_{in} \end{aligned}$$

4. For $i = 3$;

$$\begin{aligned} C_3 &= a_3 + p_3 C_2 \\ &= a_3 + p_3 (a_2 + p_2 a_1 + p_2 p_1 a_0 + p_2 p_1 p_0 \cdot C_{in}) \\ C_3 &= a_3 + p_3 a_2 + p_3 p_2 a_1 + p_3 p_2 p_1 a_0 + p_3 p_2 p_1 p_0 \cdot C_{in} \end{aligned}$$

* Circuit diagram :-

⇒ In 4-bit parallel Adder

$$S_0 = \underbrace{A_0 \oplus B_0}_{P_0} \oplus C_{in}$$

$$\begin{aligned} S_1 &= A_1 \oplus B_1 \oplus C_0 \\ &= P_1 \oplus C_0 \end{aligned}$$

$$\begin{aligned} S_2 &= A_2 \oplus B_2 \oplus C_1 \\ &= P_2 \oplus C_1 \end{aligned}$$

$$\begin{aligned} S_3 &= A_3 \oplus B_3 \oplus C_2 \\ &= P_3 \oplus C_2 \end{aligned}$$

